MILITARY SPECIFICATION

CONNECTOR, MALE, ASSEMBLY
FOR M18A1 APERS MINE

1. SCOPE

1.1 This specification covers the parts and assembly for one type of connector designated as Connector, Male, Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain for Sealing and Securing.
PPP-B-621	-	Box, Wood, Nailed and Lock Corner
PPP-B-636	-	Boxes, Fiberboard
PPP-C-843	-	Cushioning, Material, Cellulosic
UU-T-106	-	Tape, Pressure Sensitive Adhesive, Masking, Paper.

MILITARY

MIL-P-116	-	Preservation, Method of.
MIL-B-117	-	Bags, Interior Packing
MIL-C-442	-	Cable (Wire) Two Conductor, Parallel (Ripcord)
MIL-A-2550	-	Ammunition and Special Weapons; General Specification For.
MIL-F-50449-		Filler, Sheet Form (For Use in Ammunition Containers).

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables For Inspection by Attributes (ABC-STD-105)
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-1168	-	Lot Numbering of Ammunition.

FSC: 1345

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MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

9260227 - Connector, Male, Assembly
8873377 - Plug, Special

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Assembly.-The male connector assemblies shall comply with all requirements specified on Drawing (dwg.) 9260227, and with all requirements specified in applicable specifications.

3.3 Continuity and short circuit of male connector.-The male connector assembly shall be electrically continuous and shall be free of any short circuit when tested as specified in 4.4.4.

3.4 Workmanship.-All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, crazes, porosity, warpage, burn marks, blisters, excess flash, scratches, dirt, grease, rust, corrosion products and other defects and foreign matter which would affect appearance or serviceability. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat, legible and sharply defined. All packing components shall be dry.

3.5 First article inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission.-Prior to the start of regular production, the contractor shall submit a first article sample (see 6.1c) to a Government approved facility as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2 and 4.2.3. The first article sample shall consist of inert parts and sub-assemblies (see 4.2.2) and male connector assemblies (see 4.2.3) which have been produced by the contractor or furnished by a supplier and which have been

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manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Component parts and sub-assemblies.-The following component parts and sub-assemblies shall be submitted to a Government approved laboratory in accordance with the provisions of 4.2.1 for inspection as specified in 4.2.2.1:

a. Fifty (50) prongs (dwg. 9260225)
b. Twenty-five (25) prong and wire assemblies (dwg. 9260226).

4.2.2.1 Inspections to be performed.-The component parts and sub-assemblies will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

4.2.2.2 Rejection.-If any component part or sub-assembly fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any component or subassembly in the sample to comply with any of the stated requirements.

4.2.3 Male connector assemblies.-Five (5) complete male connector assemblies (dwg. 9260227), accurately identified, from each mold cavity (but not less than a total sample of 25 assemblies equally distributed from the mold cavities used) shall be submitted to a Government approved laboratory in accordance with the provisions of 4.2.1 for inspection and testing as specified in 4.2.3.1.

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4.2.3.1 Inspections to be performed.-The male connector assemblies shall be subjected by the Government to any or all of the applicable examinations and tests specified in 4.3.2 and 4.3.3 of this specification.

4.2.3.2 Rejection.-If any male connector assembly fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly in the sample to comply with any of the stated requirements.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process using the same materials and methods in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications, or drawings not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of male connector assemblies shall contain:

a. Wire from not more than one manufacturer.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4. .1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the

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contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Wire (see dwg. 9260224 covering a detail of dwg. 9260227).

Categories	Defects	Method of Inspection (see 6.4)	Code No. (see 6.3)
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Length of stripped ends	Optical Comparator	01001
102.	Wire ends not tinned	Visual	01002
Minor:	AQL 1.50 percent		
201.	Length of wire	Scale	01003
202.	Color of wire	Visual	01004
203.	Evidence of poor workmanship (see 3.5)	Visual	01005

4.3.2.2 Prong (see dwg. 9260225 covering a detail of dwg. 9260227).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Length	Snap Gage	02001
102.	Prong end length	Flush Pin Gage	02002
103.	Prong end diameter	Plain Ring Gage	02003
104.	Knurl missing or inadequate	Visual	02004
Minor:	AQL 2.50 percent		
201.	Hole diameter	Plain Plug Gage	02005
202.	Hole depth	Depth Gage	02006
203.	Body length	Height Gage	02007

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204.	Shank diameter	Plain Ring Gage	02008
205.	Loose particles in knurled area	Visual	02009
206.	Evidence of poor workmanship (see 3.5)	Visual	02010

4.3.2.3 Prong and wire assembly (see dwg. 9260226 covering a detail of dwg. 9260227).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Prong missing	Visual	03001
102.	Wire insulation damaged	Visual	03002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	03003

4.3.2.4 Male connector assembly (see dwg. 9260227)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 percent		
101.	Component improperly assembled, damaged, loose or missing	Visual	04001
102.	Depth to prong tips incorrect	Flush Pin Gage	04002
103.	Distance between centers of prongs	8873377	04003
104.	Insecure fit of dust cap	Manual	04004
105.	Presence of rubber or other foreign material on prongs	Visual	04005
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	04006

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4.3.3 Testing

4.3.3.1 Dielectric strength of wire (break down voltage) (see dwg. 9260224).-Major defect - Code No. 01006.-The sampling plan for this test shall be in accordance with MIL-STD-105 using inspection level II and an AQL of 0.40 percent. The test shall be performed as specified in 4.4.1 using equipment in accordance with 4.3.3.5 Table I. Any assembly which fails to comply with the applicable drawing requirement shall be classed defective.

4.3.3.2 Prong and wire assembly (see dwg. 9260226) - Major defect - Code No. 03004.-This pull test shall be performed 100 percent. The test shall be performed as specified in 4.4.2, using equipment in accordance with 4.3.3.5 Table I. Any assembly which fails to comply with the applicable drawing requirement shall be classed defective and removed from the lot.

4.3.3.3 Male connector assembly (see dwg. 9260227) - Major defect - Code No. 04007.-This pull test shall be performed 100 percent as specified in 4.4.3 using equipment in accordance with 4.3.3.5 Table I. Any assembly which fails to comply with the specified requirements shall be classed defective and removed from the lot.

4.3.3.4 Continuity and short circuit of male connector (see 3.3) - Major defect - Code No. 04008.-This test shall be performed 100 percent as specified in 4.4.4 using equipment in accordance with 4.3.3.5, Table I. Any assembly which fails to comply with the specified requirements shall be classed defective and removed from the lot.

4.3.3.5 Test summary.-The following table summarizes the testing required for first article and production samples. It specifies the type of inspection equipment required to perform the tests with appropriate reference to applicable notes.

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TABLE I

COMPONENT OR ASSEMBLY	TYPE OF TEST	DEFECT CODE	METHOD OF INSPECTION TEST PARA.	EQUIP.
Wire (dwg. 9260224)	Dielectric strength of wire	01006	4.4.1	CE (6.4.2)
Prong and Wire assy (dwg. 9260226)	Pull Test	03004	4.4.2	CD (6.4.1)
Male Connector Assy (dwg. 9260227)	Pull Test	04007	4.4.3	CD
Male Connector Assy (dwg. 9260227)	Short Circuit and Continuity	04008	4.4.4	CE

4.3.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed in this section is described under the "Method of Inspection" column in paragraphs 4.3.2 and 4.3.3. Special Inspection Equipment (SIE) designs, if required, are specified by drawing numbers (see 6.4.3). When no SIE design is provided, the contractor shall submit for approval inspection design in accordance with 6.4.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings provided and revised in accordance with the requirements herein may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures

4.4.1 Dielectric strength of wire (breakdown voltage).- This test shall be performed prior to assembly of the wire to the prongs. The test shall be performed as specified in MIL-C-442. Observation shall be made for any indication of breakdown.

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4.4.2 Prong and wire assembly.-Before molding into the male connector assembly, the wire shall be held securely and the required force applied to each prong in a smooth gradual manner. Observation shall be made for compliance with the applicable drawing requirement.

4.4.3 Male connector assembly.-With the male connector assembly held securely, the required force shall be applied to each prong in a smooth gradual manner. Observation shall be made for compliance with the applicable drawing requirement.

4.4.4 Continuity and short circuit.-The dust cap shall be removed and the test for short circuit made between male contacts (prongs). If no short exists, each leg of the wire shall be tested for continuity. Observation shall be made for compliance with the specified requirement.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

5.1.1 Level A.-Each wire with plug shall be folded to an approximate length of 5 1/2 inches and secured with a length of masking tape, UU-T-106, 1 inch in width. Fifty (50) plugs shall be bulk-packed in a fiber box conforming to PPP-B-636 Type CF, Class Domestic, Variety SW Grade 125 minimum, Style RSC. Box size shall be approximately 6 by 6 by 8 inches in depth inside dimensions. Cushioning material conforming to PPP-C-843, Type I or II, Class A or B shall be added as required to provide a tight pack. Closure of the boxes shall be by application of 2 inch wide gummed, kraft tape, PPP-T-45, over the full length of the top and bottom center seam of the box with an approximate 2 inch extension of tape on to the end panels of the box. All corners of the box shall be blunted and the box inserted into a barrier bag conforming to MIL-B-117 requirements for a Type I, Class E, Style 1 bag. The bag size shall be approximately 13 1/4 by 15 inches inside dimensions. All excess air shall be exhausted from the bag and the bag closed by means of heat sealing. The bag seal shall meet the heat-seal test of MIL-P-116.

5.2 Packing

5.2.1 Level A.-Twelve boxes (600 plugs) shall be packed in a wooden box conforming to PPP-B-621, Class 2, Style 4 for a Type B load. All inside surfaces of the box shall be lined with Fillers, Sheet Form, MIL-F-50449. Box size shall be approximately 26 3/4 by 12 7/8 by 12 7/8 inches in depth, inside dimensions. Side, bottom and top fillers shall be approximately 24 by 12 inches; end fillers 12 by 12 inches. Additional fillers shall be added as required to provide a

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tight pack. Boxes shall be closed and strapped in accordance with instructions contained in the Appendix of the box specification.

5.3 Marking.-Each fiber box, barrier bag and exterior container shall be marked in accordance with MIL-STD-1169 requirements.

6. NOTES

6.1 Intended use.-The male connector assembly is used to convert the M18A1 Claymore Mine from the manned to the unmanned ambush role without cutting off the female end of the Blasting Cap firing wire.

6.2 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167.
c. Provisions for submission of first article samples.
d. Serialization requirements, if applicable.
e. Quantity required and delivery schedules.
f. Inspection equipment design requirements (see 4.3.4 and 6.4).

6.3 Defect code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers and specification numbers shall be referenced on drawings of equipment designs submitted by the contractor to the Government for approval.

6.4 Inspection Equipment Designs.-The contractor shall design inspection equipment as required under the Method of Inspection column of the applicable paragraphs of Section 4. The contractor shall prepare designs for all inspections for which Special Inspection Equipment (SIE) designs have not been supplied by the Government. If SIE designs are supplied by the Government but are non-mandatory, the contractor may elect to use an approved alternate design (CD) or commercial equipment (CE). The designs shall meet the criteria described in each SIE envelope drawing supplied by the Government. Designs shall be prepared to the Category and Form (per MIL-D-1000) specified in the Contract Data Requirements List (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance.

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6.4.1 Contractor designs.-Contractor designs (designated "CD" in the Method of Inspection column of the applicable paragraphs of Section 4) are designs of inspection equipment which the Government has assigned design responsibility to the contractor, or designs of inspection equipment which the contractor has elected to prepare in lieu of non-mandatory SIE designs. Contractor designs shall include, as applicable, operating instructions, calibration procedures and maintenance procedures.

6.4.2 Commercial equipment.-Commercial equipment (designated "CE" in the Method of Inspection column of the applicable paragraphs of Section 4) is unmodified inspection equipment which is catalogued and available for purchase by the general public. Commercial equipment which the contractor proposes to use shall be fully described by means of catalog sheets or the like. This information will include illustrations and engineering data adequate for evaluation. Designs shall be prepared for any special fixtures required to be used with commercial equipment.

6.4.3 Special Inspection Equipment (SIE) designs.-An SIE design is defined herein:

a. Detailed drawings which completely depict all the information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing as defined in MIL-STD-100. This establishes the criteria which a detailed design shall meet.

6.4.4 Classification of SIE Designs

6.4.4.1 Mandatory (designated "M" in the "Method of Inspection" column of the applicable paragraphs of section 4).- The contractor shall use this design to construct equipment for the specified inspection. In the mandatory SIE design is an envelope drawing, the specified criteria shall be met in the detailed design prepared by the contractor. When the method of inspection references mandatory (M) designs, the contractor shall comply with and use these designs only.

6.4.4.2 Non-mandatory.-The contractor may use the SIE design supplied, or may design alternate inspection equipment or propose commercial inspection equipment to facilitate his operations.

6.5 Submission of contractor designs.-Contractor designs shall be approved by the Government prior to fabricating or

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procuring the equipment. Designs shall be submitted for approval within the time frame and with the distribution specified in the Contract Data Requirements List (DD Form 1423). Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt. Unless otherwise specified, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND15, Dover, New Jersey 07801. Unless otherwise specified, all other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract, submission shall be as directed by the contracting officer.

6.6 This specification is released under PAN 7100359.

Custodian:
Army-MU

Preparing Activity:
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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